

RAJALAKSHMI ENGINEERING COLLEGE
RAJALAKSHMI NAGAR, THANDALAM – 602 105



RAJALAKSHMI
ENGINEERING COLLEGE

CS23A34
USER INTERFACE AND DESIGN LAB

Laboratory Observation Notebook

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Ex. No.: 11

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Create a data visualization (e.g., pie charts, bar graphs) for an inventory management system using javascript

AIM:

The aim is to create data visualizations, such as pie charts and bar graphs, for an inventory management system using JavaScript.

PROCEDURE:

Step 1: Set Up Your HTML File

First, create an HTML file to hold your canvas for the chart and include Chart.js. html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Inventory Management Visualization</title>
  <style>    body {      fontfamily:
Arial, sans-serif;      text-align: center;
margin:
50px;
    }
```

```

        canvas {            margin:
20px auto;
        }
    </style>
</head>
<body>
    <h1>Inventory Management System</h1>
    <canvas id="pieChart" width="400" height="400"></canvas>
    <canvas id="barChart" width="400" height="400"></canvas>
    <script src="https://cdn.jsdelivr.net/npm/chart.js"></script>
    <script src="script.js"></script>
</body>
</html>

```

Step 2: Create the JavaScript File for Charts

Next, create a JavaScript file (script.js) to handle the data visualization logic.

```

javascript
// script.js

// Data for the inventory const inventoryData = {    labels:
['Electronics', 'Clothing', 'Home Appliances', 'Books', 'Toys'],    datasets:
[
    {            label: 'Items in Stock',
data: [200, 150, 100, 80, 50],
backgroundColor: [
    '#FF6384',
    '#36A2EB',

```

```

        '#FFCE56',
        '#4BC0C0',
        '#9966FF'
    ],
}
]
};

```

```

// Creating the Pie Chart const ctxPie =
document.getElementById('pieChart').getContext('2d'); const
pieChart = new Chart(ctxPie, { type: 'pie', data:
inventoryData, options: {
responsive: true, title: { display:
true, text:
'Inventory Distribution'
}
}
});

```

```

// Creating the Bar Chart const ctxBar =
document.getElementById('barChart').getContext('2d'); const barChart
= new Chart(ctxBar, { type: 'bar', data: inventoryData, options:
{ responsive: true, title: { display: true, text:
'Items in Stock by Category'
}, scales: { yAxes:
[ { ticks: {
beginAtZero: true
}

```

```
        }]  
    }  
}  
});
```

OUTPUT:

Inventory Management Dashboard

Category
All Categories

Warehouse
North

Time Period
Last Week

Update Dashboard

Total Items
8,742

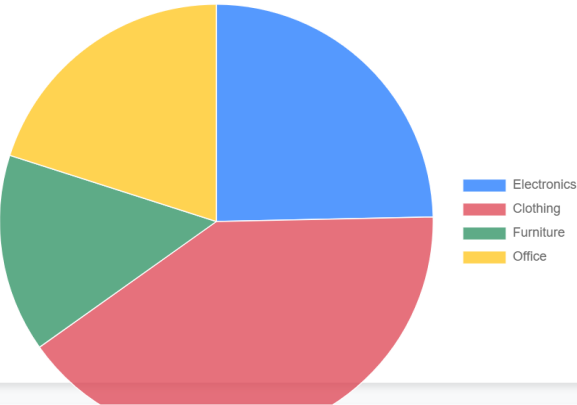
Critical Stock
28

Low Stock
156

Optimal Stock
6,213

Excess Stock
2,345

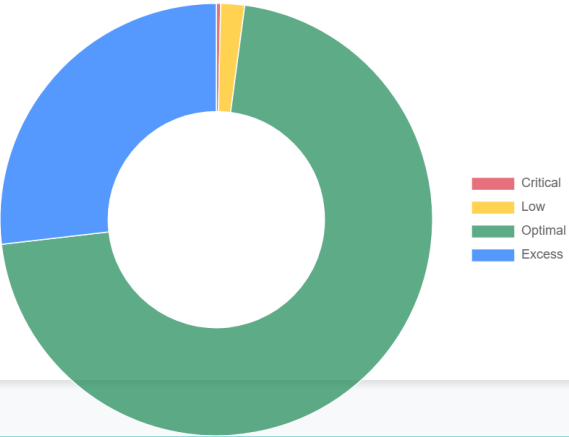
Inventory by Category



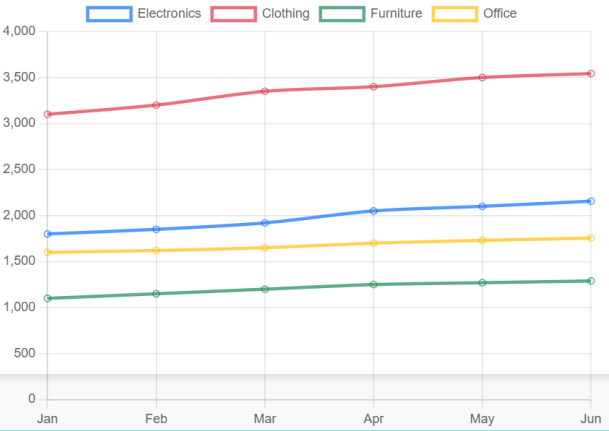
Inventory by Warehouse



Stock Level Status



Inventory Trends



RESULT:

Hence we have created data visualizations, such as pie charts and bar graphs, for an inventory management system using JavaScript